

Mobius iFlex 50 L - development design (2023/2024 evidence)

merck-iflex-50

Development-design evidence (2023/2024), not verified released-equipment specifications

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	15 L	Source-supported
Maximum working volume	50 L	Source-supported
Nominal vessel / bag size	50 L	Source-supported
Total vessel volume loaded into the form	50 L	Derived / estimated
Initial working volume loaded into the form	15 L	Derived / estimated
Product-contact material	Plastic / single-use	App assumption
Height : diameter	1.62	Derived / estimated
Internal / chamber diameter	0.34 m	Source-supported
Liquid height at maximum working volume	0.5507 m	Derived / estimated
Impeller / mixing type	Pitched-blade turbine	Source-supported
Impeller count	1	App assumption
Impeller diameter	0.117 m	Source-supported
Power number	3.6	Source-supported
Initial agitation	100 rpm	App assumption
Maximum agitation	250 rpm	App assumption
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.3 vvm	App assumption
Baffled vessel	Yes	Source-supported
Maximum pressure	0.5 bar(g)	App assumption
Default operating pressure	0.05 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 15 L

Source-supported | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1, 50 L column (asterisk: in development)

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 15 L.

Limitations: DEVELOPMENT DESIGN: source table marks this size in development. These are manufacturer characterization-design values, not confirmed released-equipment specifications. No transfer from the older non-iFlex Mobius family. Four-blade down-pumping pitched impeller; X-baffle; bottom mounted 15 degrees from centre.

Local original: [Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 \(July 2023\) - PDF page 2](#)

Maximum working volume - 50 L

Source-supported | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1, 50 L column (asterisk: in development)

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 50 L.

Limitations: DEVELOPMENT DESIGN: source table marks this size in development. These are manufacturer characterization-design values, not confirmed released-equipment specifications. No transfer from the older non-iFlex Mobius family. Four-blade down-pumping pitched impeller; X-baffle; bottom mounted 15 degrees from centre.

Local original: [Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 \(July 2023\) - PDF page 2](#)

Nominal vessel / bag size - 50 L

Source-supported | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1, 50 L column (asterisk: in development)

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 50 L.

Limitations: DEVELOPMENT DESIGN: source table marks this size in development. These are manufacturer characterization-design values, not confirmed released-equipment specifications. No transfer from the older non-iFlex Mobius family. Four-blade down-pumping pitched impeller; X-baffle; bottom mounted 15 degrees from centre.

Local original: [Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 \(July 2023\) - PDF page 2](#)

Total vessel volume loaded into the form - 50 L

Derived / estimated | Form-population rule

Rule: total volume = max(nominal vessel volume, maximum working volume)

Selection / calculation: $\max(50 \text{ L}, 50 \text{ L}) = 50 \text{ L}$

Limitations: This prevents the form's total vessel volume from being smaller than the working-volume ceiling.

Initial working volume loaded into the form - 15 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: $\min(15 \text{ L}, 50 \text{ L}) = 15 \text{ L}$

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Plastic / single-use

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Plastic / single-use.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Height : diameter - 1.62

Derived / estimated | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1, 50 L column (asterisk: in development)

Rule: $H = 4 V / (\pi D^2)$; liquid $H/D = H / D$

Selection / calculation: $V = 0.05 \text{ m}^3$; $D = 0.34 \text{ m}$; $H = 0.5507091456466966 \text{ m}$; $H/D = 1.6197327813138134$

Limitations: Derived liquid aspect ratio. Does not reuse an overall vessel ratio or generic profile ratio as if it described the liquid.

Local original: Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2

Internal / chamber diameter - 0.34 m

Source-supported | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1, 50 L column (asterisk: in development)

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.34 m.

Limitations: DEVELOPMENT DESIGN: source table marks this size in development. These are manufacturer characterization-design values, not confirmed released-equipment specifications. No transfer from the older non-iFlex Mobius family. Four-blade down-pumping pitched impeller; X-baffle; bottom mounted 15 degrees from centre.

Local original: Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2

Liquid height at maximum working volume - 0.5507 m

Derived / estimated | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1, 50 L column (asterisk: in development)

Rule: $H = 4 V / (\pi D^2)$; liquid $H/D = H / D$

Selection / calculation: $V = 0.05 \text{ m}^3$; $D = 0.34 \text{ m}$; $H = 0.5507091456466966 \text{ m}$; $H/D = 1.6197327813138134$

Limitations: Ideal-cylinder estimate from published diameter and working volume. Ignores tapered walls, curved bottoms and internals; not a manufacturer liquid-height specification.

Local original: Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2

Impeller / mixing type - Pitched-blade turbine

Source-supported | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1, 50 L column (asterisk: in development)

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Pitched-blade turbine.

Limitations: DEVELOPMENT DESIGN: source table marks this size in development. These are manufacturer characterization-design values, not confirmed released-equipment specifications. No transfer from the older non-iFlex Mobius family. Four-blade down-pumping pitched impeller; X-baffle; bottom mounted 15 degrees from centre.

Local original: Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2

Impeller count - 1

App assumption | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1 and Figure 1

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 1.

Limitations: Single bottom-mounted design interpreted from Table 1 and Figure 1; not an explicitly tabulated impeller count. Development-design qualification applies.

Local original: Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2

Impeller diameter - 0.117 m

Source-supported | Merck Mobius iFlex Scalability and Performance Guide - PG12163EN Rev. 2 (March 2024) - PDF page 4; Table 1, 50 L column, Impeller Diameter (cm)

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.117 m.

Limitations: DEVELOPMENT DESIGN: source table marks this size in development. These are manufacturer characterization-design values, not confirmed released-equipment specifications. No transfer from the older non-iFlex Mobius family. Direct absolute dimension from March 2024 guide; preferred to multiplying rounded di:D ratios from the application note.

Local original: [Merck Mobius iFlex Scalability and Performance Guide - PG12163EN Rev. 2 \(March 2024\) - PDF page 4](#)
Manufacturer publication

Power number - 3.6

Source-supported | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1, 50 L column (asterisk: in development)

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 3.6.

Limitations: DEVELOPMENT DESIGN: source table marks this size in development. These are manufacturer characterization-design values, not confirmed released-equipment specifications. No transfer from the older non-iFlex Mobius family. Four-blade down-pumping pitched impeller; X-baffle; bottom mounted 15 degrees from centre.

Local original: [Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 \(July 2023\) - PDF page 2](#)

Initial agitation - 100 rpm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baseRpm = 100`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial agitation setting loaded into the form.

Maximum agitation - 250 rpm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxRpm = 250`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the ceiling available to the generic DO-control cascade.

Initial aeration - 0.05 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].baseVvm = 0.05`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.3 vvm

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxVvm = 0.3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - Yes

Source-supported | Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023) - PDF page 2; Table 1, 50 L column (asterisk: in development)

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Yes.

Limitations: DEVELOPMENT DESIGN: source table marks this size in development. These are manufacturer characterization-design values, not confirmed released-equipment specifications. No transfer from the older non-iFlex Mobius family. Four-blade down-pumping pitched impeller; X-baffle; bottom mounted 15 degrees from centre.

Local original: [Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 \(July 2023\) - PDF page 2](#)

Maximum pressure - 0.5 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].maxPressure = 0.5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0.05 bar(g)

App assumption | Generic mammalian stirred-tank profile

Rule: `PROFILE_DEFAULTS["mammalian"].operatingPressure = 0.05`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable mammalian stirred-tank starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: `maxOxygenFraction = 100`

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

These presets are primarily mammalian-cell-culture archetypes.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

DEVELOPMENT DESIGN: source table marks this size in development. These are manufacturer characterization-design values, not confirmed released-equipment specifications. No transfer from the older non-iFlex Mobius family.

AN12431EN p. 4 RPM/PV pairs are characterization points, not motor-speed ratings. Existing RPM and pressure ceilings remain unverified assumptions. Absolute impeller diameters are from PG12163EN Rev. 2 p. 4; liquid heights remain cylinder estimates.

Geometry caution: AN12431EN p. 2 prints minimum/maximum liquid-height ratios greater than one; PG12163EN p. 4 repeats this. A reversed ratio heading is plausible but unconfirmed. The maximum-fluid/total-height ratio 0.8 is nominal, not an absolute liquid-height measurement. Neither ratio overwrites the existing liquid-height calculation. Validation scope and conflicts: [sources/AN12431-REVIEW.md](#).

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Merck Mobius 3 L to iFlex scalability - AN12431EN Ver. 1.0 (July 2023)

[merck-an12431-v1-2023-9372dd222c24.pdf](#)

SHA-256: 9372dd222c246b5b420a883bb89f5132e68dacbbea4e32164dfc0fbb48f41aeb

Merck Mobius iFlex Scalability and Performance Guide - PG12163EN Rev. 2 (March 2024)

merck-iflex-pg12163-rev2-2024-bc57dac53baf.pdf

SHA-256: bc57dac53baf1e19c190fdda59e7adb8d2cfa430865caa5bfad77d98119e652b